

CAREER SUMMARY:

Mechanical Research and Development Engineer and automotive enthusiast with 10+ years of experience in mechanical CAD design, including ANSYS FEA and CFD, engineering design and manufacturing/machining, and injection molding and sand casting. Listed for both significant and primary engineering contributions on more than 10 published utility and design patents. Currently holding a Aerospace Design Engineer position with extensive experience in DPAS, ITAR and EAR, turbocharger testing and design, and high precision manufacturing.

LIST OF QUALIFICATIONS

Expert Proficiency in Mechanism and Part Design, Fabrication, and Presentation for Advanced New Product Development.
Expert Proficiency with ANSI Standards and ASME Y14.5, and GD&T 2018.
Expert Proficiency with SolidWorks, Creo, KeyShot, AutoCAD, Fusion360, QuickSurface Pro, and DesignX.
Expert Proficiency with 3D scanning and (Zeiss/GOM + Inspect, Creaform Metrascan, and Revopoint MetroY Pro).
Very proficient with SLA, FDM, SLS and DMLS rapid prototyping techniques and concept product development.
Very proficient with Sheet Metal, EDM, and Machining Processes for production manufacturing and VA/VE.
Very proficient with UL/(c)UL, ETL, IFETEL, FCC, RoHS and IP standards and design.
Skilled with TIG and MIG (mild steel and stainless-steel) welding.
Familiar with KiCad and Altium Designer for Electrical systems and schematics.
Familiar with ANSYS mechanical FEA and BaramFlow CFD setup and post-processing.
Familiar with VBA programming, Allen-Bradley PLC, and G-Code.
Six Sigma Green Belt Certified (Garrett Training).

EXPERIENCE

Aerodyn Engineering LLC; Aerospace / Defense Engineering and Analysis, Turbocharger and Jet Turbine Engine Instrumentation
Design Engineer – Current Role

- Design, analyze and provide manufacturing support for flight test hardware, turbocharger test hardware, and high tolerance machining fixtures.
- Complied with MIL and AS standards and corporate engineering standards from Blue Origin, Rolls-Royce, Garrett, SpaceX, GE Aerospace for bespoke design and drafting.
- Write and orally present whitepaper topics on membrane stress hand calculations for pressure vessels
- Program management and schedule management (Msoft Project 2016) for large frame axial flow turbine, serial compressor turbocharger design and assembly.
- Provide high tolerance 3D scanning support for QC and reverse engineering.
- Utilize ANSYS 2024b for static structural and modal analysis of on engine instrumentation and turbine test setups.

Garrett Advancing Motion (NYSE:GTX); Commercial, OEM, and Aftermarket Automotive Turbocharger Products

Product Development Engineer II, August 2022 – August 2024

- Use thermodynamic, kinematic, and statistical tools to simulate turbocharger performance specifications to meet customer requirements
- Supported SMEs and CAD designers in creating and testing bespoke turbine housing and compressor housing design, casting, and machining
- Work with tier 1 manufacturers like Caterpillar, John Deere, and Ford Motor Company to bring product plans from phase 0 concept design to final product delivery
- Travelled internationally to support prototype turbocharger development and pioneered the CV north America RAIL meeting process.

Holley Performance Products Inc. (NYSE:HLLY); Automotive Aftermarket and Performance Parts

Lead Mechanical Engineer, January 2021 – August 2022

- Worked as the primary mechanical engineer and CAD designer for Ford 7.3L performance sand cast exhaust manifolds
- Lead the NOS Nitrous and Earl's Performance Fittings departments designing new products and supporting existing lines
- Worked with tier 1 and other aftermarket manufacturers to create new nitrous and fuel delivery products

Rev-a-Shelf; Retail and OEM Kitchen and Bath Organizers

New Product Development Engineer, September 2019 – January 2021

- Lead the conceptual and mechanical design refresh for a new generation of organizers and waste containers
- Created photorealistic scenes and animations to debut concept products at KBIS 2019
- Created and presented Creo 5.0 FEA simulations to prove aluminum structural pieces before tooling

HeathCo, LLC; Security and Decorative Lighting and Chimes

Mechanical NPD Engineer I, May 2017 – September 2019

- Eliminated over \$40,000 of yearly departmental spending by transitioning the prototyping operation in-house

LEEMAN E. STEVENS

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- Created department-wide Excel VBA automated programs and utilities for sustaining engineering and document control
- Analyzed mechanical design and manufacturing processes to eliminate more than 15% of manufacturing costs across 20 SKUs
- Pioneered thermal simulation for proving heatsink design for high powered LED (LV/LE and 120v+, 2000+ lumen)

SpanTech, LLC; Custom Factory Automation Design and Manufacturing

R&D Engineer, Design Engineer, May 2014 – May 2017

- Created prototype and production soft jaws, components, and assemblies for custom automation and material handling projects
- Created automated SolidWorks scripts and assemblies to standardize difficult design hurdles
- Awarded for creating and developing Interactive Engineering Online Training Center

EDUCATION

Murray State University – Institute of Engineering

Bachelor of Science in Design Engineering Technology (Mechanical / Manufacturing) – 2016 – 3.33/4.00

REFERENCES AVAILABLE UPON REQUEST